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Mesh-Free Dexterous Robot Demonstrations from Human RGB-D Video and Teleoperation

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Abstract

Teleoperating a five-fingered robot hand to collect demonstrations is costly and scales poorly, while human video is abundant yet contains neither robot commands nor robot images. Existing approaches overlay rendered robots on human video, mainly for parallel-jaw grippers; reconstruct object meshes or point clouds in place of the original RGB; or require a wearable exoskeleton during capture. We present a pipeline that composes stationary RGB-D video of a bare human hand with teleoperation logs of the same task. The video supplies the metric hand trajectory and the scene; the teleoperation log supplies recorded finger position targets. Persistent hand-state transition alignment (PHTA) matches ordered changes in a common hand-shape representation and transfers the complete finger-command vector through one temporal correspondence, without task-phase, contact, or grasp labels. Part-aware visibility transfer replaces the human hand with the robot hand while preserving which parts were in front of or behind the object, requiring no object mesh or pose. Combining N videos with M demonstrations yields $N \times M$ demonstrations of teleoperation quality. On four real-robot tasks, five teleoperation demonstrations composed with twenty human episodes raise behavior-cloning success from 0–45% to 50–90%, matching one hundred teleoperation demonstrations on two tasks.

Keywords: learning from demonstration; behavior cloning; dexterous manipulation; human video; teleoperation; hand pose estimation; RGB-D perception; temporal alignment; video inpainting; robot data generation

1. Introduction

A manipulation policy is learned from demonstrations, and the standard way to obtain them is teleoperation. An operator drives the robot through the task, and every recorded frame is a robot command paired with an image of the robot performing it. The data is exactly what a policy needs, and it does not scale: each demonstration costs an operator, a robot, and real time, and that cost grows linearly with the size of the dataset.

Video of people doing the same tasks has the opposite profile. It is abundant, cheap to record, and needs no robot. What it does not contain is a robot—neither the commands a robot would execute nor an image of a robot executing them. Turning such video into usable demonstrations therefore means closing two gaps at once. The *executability gap* is that a human hand pose is not a robot command: the two hands differ in joint count, link

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length, and joint axis, so a pose transcribed directly produces a posture the robot cannot reach. The *visual gap* is that the recorded frames show a human hand, while the policy will act from frames showing a robot hand.

Existing approaches close these gaps in three broad ways, each with a cost. *Co-training* mixes human and robot data and relies on a shared representation to transfer; it leaves both gaps open and uses the robot data only to anchor a learned transfer, rather than to supply the executed commands. *Reconstruction* recovers object geometry and three-dimensional hand–object trajectories from the video and trains a policy on that reconstructed state [1,2]; this discards the original RGB, and some of these methods additionally reconstruct an object mesh from the video [1]. *Device-based* capture has the person wear an exoskeleton or hold an instrumented interface so the recorded motion is robot-feasible by construction [3]; this closes both gaps but restores hardware at capture time, which is precisely what recording human video was meant to avoid.

We take a different position, and it starts from an asymmetry between the two gaps. The visual gap concerns appearance and can be repaired in image space: the human hand can be removed and a robot hand drawn where it was. The executability gap concerns physics and cannot be repaired that way—no amount of image editing makes an unreachable finger posture reachable. We therefore take the finger position targets from successful teleoperation logs and use human video to supply the metric hand-root trajectory and visual context of the scene. The recorded targets include control adjustments that cannot be recovered from visible human posture alone.

Composing two independently recorded sources requires a correspondence between their movements. In time, we compare hand-shape transitions in a common dimensionless representation, together with normalized translation progress, rather than directly equating native joint angles or absolute workspace coordinates. The resulting correspondence determines when to use the recorded robot finger commands. In image space, we read from the human video which hand parts were in front of the manipulated object and which were behind it, and apply that relation to the corresponding robot parts.

Because the video and the teleoperation log are recordings of different trials, they can be recombined freely. N human episodes and M teleoperation demonstrations yield $N \times M$ robot demonstrations, and the expensive perception stages run only N times. Adding a demonstration costs a composition, not a capture. Mixing inexpensive human video into a few teleoperation demonstrations in this way yields training data of teleoperation quality. In real-robot experiments on four tasks, five teleoperation demonstrations composed with twenty human episodes train policies that succeed in 50–90% of the trials, whereas the same five demonstrations alone reach at most 45%; on two tasks the composed data match one hundred teleoperation demonstrations.

One term in the title needs its scope fixed at the outset. *Mesh-free* here refers to the object: no stage of the pipeline requires an object computer-aided design (CAD) model or a six-degree-of-freedom object pose, and a new object is introduced by a text prompt. It does not mean the method is free of meshes altogether. We use the MANO parametric hand model [4], and its 778 vertices carry the dense depth term of Section 3.2; the robot hand is rendered from its own mesh assets. The claim is about what the user must supply, not about what the system draws.

The contributions of this work are as follows:

- **Role-separated composition.** A pipeline that combines scene appearance and metric hand-root motion from human RGB-D video with recorded robot finger position targets from teleoperation, and reuses the independently recorded sources across compatible pairings.

- **PHTA.** An alignment method that associates ordered hand-state transitions using multiscale deformation evidence, calibrated direction compatibility, and intrinsic translation-progress context. It determines the target event inventory from data and applies one temporal correspondence to the complete finger-command vector, without annotated task phases, contact states, or grasp labels.
- **Part-aware visibility transfer.** A compositing rule that recovers hand–object occlusion from the human video and transfers it to the robot hand through a semantic part correspondence, requiring neither object mesh nor object pose.

Together, these components turn a small number of teleoperation demonstrations and inexpensive human video into teleoperation-level training data, which we verify with behavior cloning on a physical robot.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 presents the method and the experimental protocol. Section 4 reports the results. Section 5 discusses them together with the limitations of the present system, and Section 6 concludes.

2. Related Work

2.1. Learning Manipulation from Human Video

One line of work reconstructs the scene. Object geometry and three-dimensional hand–object trajectories are recovered from RGB video and a policy is trained on that reconstructed representation [1]. A recent large-scale variant learns universal dexterous control from egocentric human video by attaching a robot hand to a reconstructed point cloud [2]. These methods obtain a state a robot can be driven toward, but the appearance of the original scene is not preserved, the policy class is tied to the reconstructed representation, and some methods additionally depend on a reconstructed object mesh [1]. Our pipeline keeps the original RGB as the observation and requires no object model.

Human data has also been used directly for multi-fingered hands. DexMV estimates hand and object poses from video, retargets the human motion to a robot hand in simulation, and uses the result for imitation learning [5]. VideoDex pretrains policies with visual, action, and physical priors extracted from internet videos of human hands [6], and DexCap records hand motion with a portable motion-capture glove and retargets it to the robot by inverse kinematics [7]. In all three the robot finger motion is derived from the human hand. We avoid that step: the finger commands are recorded on the robot itself.

A second line transfers at the level of representation, training on human and robot data together or mapping between the two domains [8,9]. This is complementary to our work: it changes how a policy consumes data, whereas we change what data exists. Our output is an ordinary paired dataset of images and joint commands and imposes no policy architecture.

2.2. Cross-Embodiment Visual Transfer

The closest work edits the video itself, removing the human and drawing a robot in its place. One group has developed this line for parallel-jaw grippers, first training policies from edited human video alone [10] and then extending the approach to in-the-wild footage [11]. Related systems render photorealistic robot overlays for zero-shot imitation [12], or synthesize robot video generatively rather than by compositing [13], and low-cost variants combine edited human video with a small amount of teleoperation [14]. Image editing has also been used between robots, synthesizing demonstrations of one robot from those of another to transfer policies across embodiments [15].

Two things separate our setting. The first is the end effector: a parallel-jaw gripper has one degree of freedom, so a plausible gripper state can be inferred from a human grasp,

whereas a five-fingered hand cannot be filled in this way. The authors of the gripper-based systems note this boundary themselves: demonstrations were restricted to pinch grasps because the robots used parallel-jaw grippers [10], and retargeting dexterous human grasps to a parallel-jaw gripper was described as imperfect, with dexterous end effectors identified as a way to narrow the embodiment gap [11]. The second is what supplies the action. These methods infer the robot's action from the human motion; we take the finger commands from teleoperation and infer only the hand root from video. A generative renderer additionally produces frames without a guarantee that the depicted motion is executable, while our compositing is deterministic and every rendered frame corresponds to a command that satisfies the robot's velocity limits.

A third approach removes the embodiment gap at capture time by having the demonstrator wear an exoskeleton shaped like the robot hand, and inpaints the robot hand over the recorded human hand [3]. This yields directly-executable dexterous demonstrations, at the cost of a wearable device and therefore of any recording made without one. We require no device on the demonstrator.

2.3. Retargeting and Components

Mapping human hand motion onto a dexterous robot hand is a well-studied problem, and vision-based teleoperation systems solve it online across many hand morphologies [16]. Our use of retargeting is narrower: we retarget only the hand root, because the finger degrees of freedom are supplied by the teleoperation log rather than estimated.

The pipeline builds on established components. Hand trajectory recovery follows DynHaMR [17], a multi-stage optimization that recovers global hand motion from monocular video of a moving camera by combining a per-frame transformer-based reconstructor [18] over the MANO model [4], camera motion from dense visual simultaneous localization and mapping (SLAM) [19], and a learned hand motion prior [20]. Our setting differs in three respects. The camera is stationary, so the camera estimation is removed; depth is measured, so metric scale is taken from the depth term of Section 3.2 rather than from the monocular reconstruction; and the hand shape is held fixed while the motion prior is not used. Visible hand masks come from a promptable segmentation model [21,22], amodal object silhouettes from a diffusion-based video amodal segmentation model [23], and the clean plate from video inpainting [24]. A mesh-based pose tracker [25] is supported as an alternative mask backend for objects with a registered model, but it is not used in the configuration reported here.

The temporal alignment draws on standard statistical and geometric tools rather than a task-specific heuristic: scale-space analysis for multi-scale event detection [26], an accumulated-energy merge rule inspired by topological persistence [27], model selection by BIC [28], directional statistics on the sphere [29], and isotonic regression for monotone progress [30]. Classical dynamic time warping [31] provides the pairwise baseline against which we compare. Differentiable relaxations of dynamic time warping [32] and self-supervised video alignment by temporal cycle consistency [33] align sequences through a shared or learned feature space of one modality. PHTA aligns two embodiments instead, through a dimensionless hand state that both express, and requires no training.

Unpaired data by itself is not new: generative translation can be trained without paired human–robot videos [13], and co-training mixes edited human video with separately collected robot demonstrations [11,14]. Producing many demonstrations from few is likewise the aim of MimicGen, which adapts a small number of robot demonstrations to new object configurations in simulation [34]; our recombination operates on real video and needs no simulator state of the objects. What differs is where the pairing is made. We construct each demonstration by composing an unpaired human episode with an unpaired

teleoperation demonstration, so that the images come from the video and the commands come from teleoperation, and the dataset grows with the product of the two sources rather than with either alone.

3. Materials and Methods

3.1. Problem Setup and Overview

The system consumes two recordings of the same manipulation task. The two are captured independently and never at the same time.

The first is a *human episode* V: a stationary RGB-D recording of a person performing the task with a bare hand. It supplies two things a teleoperation log cannot. It gives the metric three-dimensional path that the hand travels through the real scene, and it gives the visual context of the task—the actual objects, where they were placed, and how they appear under the room's lighting.

The second is a *teleoperation bundle* T: a log recorded while an operator drove the robot through the same task. Each demonstration carries timestamps, six right-arm joint positions, and fifteen right-hand position-target commands. The finger targets can differ from realized joint positions during contact; they are the control quantities transferred by our method.

The two recordings supply complementary information: robot-native finger commands from teleoperation, and the scene and hand-root trajectory from the human episode.

We compose the two into image–command pairs $\{(I_k, q_k)\}$, where I_k is the composed image and q_k is its associated robot command. The finger-command path is supplied by T (Section 3.4). The six-degree-of-freedom hand-root trajectory is obtained by anchoring the robot fingertips, at the transferred finger posture, to the human fingertips recovered from V (Section 3.3). Hand geometry and normalized translation progress determine the temporal correspondence. The image is built from the corresponding frame of V with the human hand replaced (Section 3.6). Finger targets are interpolated and resampled along the recorded command path.

The hand-root trajectory and finger-command path retain their respective sources. PHTA supplies their temporal correspondence, and the output schedule specifies how quickly the composed path is traversed. Figures 1 and 2 summarize the inputs and pipeline.



(a) Human RGB-D recording (b) Composed demonstration

Figure 1. Input and output. (**a**) A frame of the human RGB-D recording. (**b**) The composed frame, in which the robot arm and hand occupy the place of the human arm and hand. The hand root follows the video; the finger commands come from an independently recorded teleoperation demonstration.

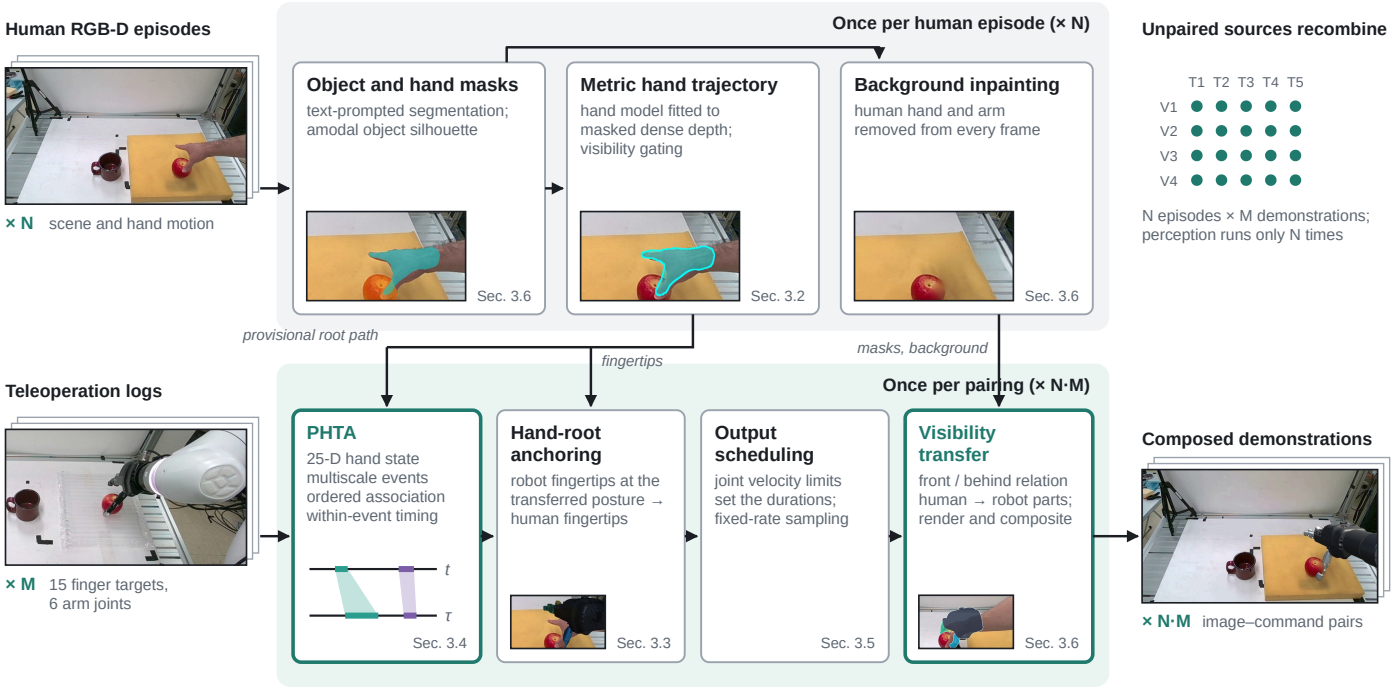


Figure 2. Pipeline overview. Stages that depend only on the human episode—object and hand masks, hand trajectory recovery, and background inpainting—are computed once per episode. Stages that depend on a pairing—temporal alignment, fingertip-anchored hand-root placement, output scheduling, and visibility-aware rendering—are computed once per (episode, demonstration) combination. N human episodes and M teleoperation demonstrations therefore yield $N \times M$ demonstrations while the perception stages run only N times. Blocks with a colored border are contributions of this work; section numbers refer to the text.

3.1.1. Formal Statement

Write the human episode as T_S frames and the teleoperation demonstration as T_T samples. Composition requires a correspondence from teleoperation index to source time,

$$\begin{aligned} &\varphi : [0,\, T_T - 1] \to [0,\, T_S - 1], \\ &\varphi(0) = 0, \qquad \varphi(T_T - 1) = T_S - 1, \\ &\varphi(\tau_1) \le \varphi(\tau_2) \quad \text{for } \tau_1 < \tau_2, \end{aligned} \tag{1}$$

which is monotone so that the order of task states is preserved and complete so that neither recording is truncated. Given φ, with $b_V(\cdot)$ the hand root pose on the source timeline, obtained from the human fingertips of the video and the transferred finger posture (Section 3.3), and $u_T(\cdot)$ the recorded finger command, the composed robot state at teleoperation index k is

$$\hat{b}_k = \mathrm{Interp}\big(b_V,\, \varphi(k)\big), \quad \hat{u}_k = u_T(k), \quad \hat{q}_k = \mathrm{Assemble}\big(\hat{b}_k,\, \hat{u}_k\big). \tag{2}$$

Note that $\varphi(k)$ is a continuous source time, not an integer frame index. The hand root is evaluated at that continuous time, while the image observation is drawn from the nearest recorded frame; the correspondence is not defined by the image sampling. Sections 3.4 and 3.5 construct φ and then convert it into an executable schedule.

3.1.2. Input Data

A human episode stores a color video, one depth image per color frame registered to the color pixels, and the camera intrinsics. When the stored frame is a crop of a larger sensor image, the principal point is shifted by the crop origin, so that every projection uses the stored image size and intrinsics. Table 1 summarizes the notation.

Table 1. Notation.

Symbol	Meaning
V, T	human RGB-D episode; teleoperation bundle
T_S, T_T	number of source frames; number of teleoperation samples
t, τ	source (video) time; teleoperation time
φ	monotone correspondence $t = \varphi(\tau)$ from teleoperation to source time
$b_V(\cdot)$, $u_T(\cdot)$	hand-root pose on the source timeline; recorded finger command
$\hat{q}_k$	composed robot state at teleoperation index k
Ψ	admissible vertex set of the dense depth term
$x(t)$	25-dimensional intrinsic hand state
$\Delta_h(n_0)$	transition deformation at half-width h and grid center n_0
E_b, v_b	accumulated energy and representative direction of branch b
K	number of persistent target events
ρ_i	posterior that source branch i belongs to the high-energy component
κ	calibrated von Mises–Fisher concentration of direction agreement
z	normalized hand-root translation progress, $z \in [0, 1]$
$g_i^S(z)$, $g_j^T(z)$, Γ_{ij}	event densities over progress; their overlap
S_{ij}	association score of source event i and target event j
γ_n	responsibility of the localized timing component within a source cell
$H_{t,p}$, M_t^H, O_t	part projection; visible human hand; object silhouette
F, B	front and behind regions
χ	human-to-robot semantic part correspondence
M_t^R	visible robot mask
N, M	numbers of human episodes and teleoperation demonstrations

3.1.3. Reuse Boundary and Combinatorial Scale

The stages that depend only on V—object masks, visible hand masks, hand trajectory recovery, the provisional hand-root path, and background inpainting—are computed once per human episode. The stages that depend on a pairing—temporal alignment, fingertip-anchored hand-root placement, inverse kinematics, output scheduling, and rendering—are computed once per (episode, demonstration) combination. The hand-root placement is a closed-form solution per frame. Because the two sources are unpaired, N human episodes and M teleoperation demonstrations yield $N \times M$ robot demonstrations while the expensive perception stages run only N times. The cost of adding a demonstration is therefore the cost of composition, not the cost of perception (Figure 2).

3.2. Metric-Scale Hand Trajectory from RGB-D

3.2.1. Why Metric Scale Is Required

Hand reconstruction from monocular video recovers a trajectory that is correct only up to an unknown scale and an unknown distance from the camera. That is sufficient when the goal is to describe a motion, but not when a robot must place its own hand where the human hand was: an error in scale becomes an error in reach. We therefore recover the hand trajectory in metric units from RGB-D, and treat the measured depth as the primary constraint on where the hand is.

3.2.2. Optimization

The optimization follows the multi-stage structure of Dyn-HaMR [17]. Per-frame hand pose and shape are initialized with a transformer hand reconstructor [18] and then refined over the whole episode by root fitting followed by temporal smoothing, with the hand shape held fixed (Section 3.2.5). The camera is stationary, so no camera motion is estimated and the whole trajectory is expressed in the one fixed camera frame. Two-dimensional keypoint reprojection anchors the optimization in the image, and the depth term described next anchors it in metric space.

3.2.3. Depth Supervision

The hand model has about sixty parameters once articulation, shape, and global pose are counted together. Supervising depth at a small number of joints is therefore weakly constraining: the residual is absorbed by articulation rather than by the quantity of interest, which is where the hand is. The optimizer supports three depth terms of increasing spatial coverage (Figure 3), of which the reported configuration uses the third:

- *Wrist depth.* A single point, and the point least affected by self-occlusion. It constrains the coarse question of where the hand is.
- *Fingertip depth.* Five points that matter most for contact, but also the points most often occluded.
- *Masked dense depth.* Rather than asking where a few landmarks are, this term asks whether the visible hand surface looks the way the depth sensor measured it.

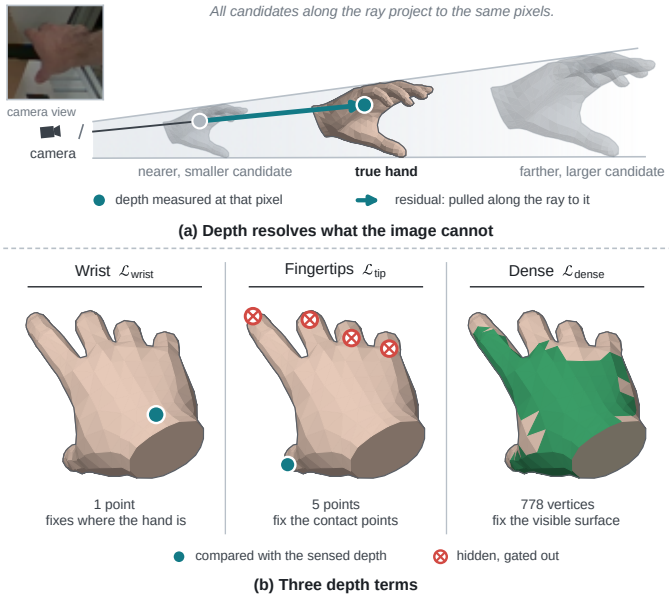


Figure 3. Depth supervision. (**a**) Hands of different size at different distances along one camera ray project to the same pixels, so the image alone does not fix metric scale; the depth measured at that pixel supplies a residual along the ray. (**b**) The depth terms supported by the optimizer: the wrist term $\mathcal{L}_{\text{wrist}}$ constrains the coarse position with one point, the fingertip term $\mathcal{L}_{\text{tip}}$ constrains the five contact points and drops those that are hidden, and the masked dense term $\mathcal{L}_{\text{dense}}$ compares the visible part of the 778-vertex hand surface with the measured depth image. The reported configuration uses $\mathcal{L}_{\text{dense}}$ (Table 2).

The dense term projects the 778 vertices of the hand mesh into the image, reads the measured depth at those pixels, and penalizes the difference. Writing $\pi(v)$ for the projection of vertex v, d_v for its predicted camera-space depth and $\hat{d}$ for the measured depth map,

$$\mathcal{L}_{\text{dense}} = \frac{1}{|\Psi|} \sum_{v \in \Psi} \text{SmoothL1}_{\beta}\Big(d_v - \hat{d}\big(\pi(v)\big)\Big), \qquad \beta = 5\,\text{mm}. \tag{3}$$

Here $\text{SmoothL1}_{\beta}(e) = e^2/(2\beta)$ for $|e| < \beta$ and $|e| - \beta/2$ otherwise, with e in meters; this is the Huber loss divided by β, the scale to which the weights of Table 2 refer. It is a two-and-a-half-dimensional reprojection: the comparison is made against a measured depth image, not against a reconstructed three-dimensional surface. The depth map is read by nearest-neighbour sampling rather than bilinear interpolation, because a depth image has hard edges between the near hand and the far background and blending across such an edge creates gradients that pull vertices into empty space.

The admissible set Ψ retains a vertex only when every one of the following holds: it projects inside the image; the measured depth there is valid; the sensor confidence at that pixel is at least 0.8; the pixel is not covered by the visible object mask; the frame is marked visible; the vertex faces the camera; it passes the self-occlusion test described below; and the residual is below 100 mm. The last condition is evaluated without gradient and prevents the optimizer from diverging in order to fit a vertex onto a background pixel.

The reported configuration uses the masked dense term alone, together with the fixed hand shape of Section 3.2.5 and two-dimensional keypoint reprojection; the wrist and fingertip terms remain available in the optimizer but are disabled. The weights are listed in Table 2.

Table 2. Weights of the depth terms over the optimization stages in the reported configuration.

Term	Root fitting	Smoothing
Wrist joint depth	0	0
Fingertip depth	0	0
Masked dense depth	50,000	50,000

The wrist and fingertip terms are supported by the optimizer but disabled in the reported configuration.

3.2.4. Visibility Gating

A depth measurement constrains a mesh vertex only if that vertex is the surface the sensor actually saw. Two things break this. The manipulated object can stand in front of the hand, and the hand can stand in front of itself. Object occlusion is resolved with the visible object mask, the modal mask from which the amodal silhouette of Section 3.6 is completed. Self-occlusion is resolved by an always-on geometric test (Figure 4). A vertex v is rejected when another vertex w of the same hand satisfies both of the following conditions:

- it projects into the same six-pixel screen cell as v, so that it plausibly blocks the line of sight;
- it is closer to the camera than v by more than 15 mm, so that only surfaces that are definitely in front are treated as occluders.

The depth margin also keeps the skin immediately surrounding v from counting as its own occluder, because that skin differs from v in depth by less than 15 mm.

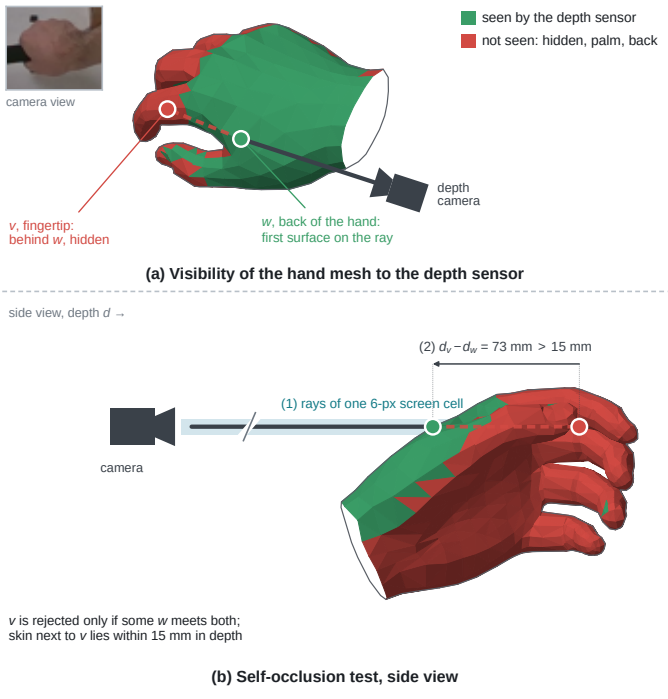


Figure 4. Self-occlusion gating; occlusion by the object is handled by the object mask. (**a**) Hand-mesh vertices seen by the depth sensor (green) and not seen (red) in one frame; the fingertip vertex v lies behind the back-of-hand vertex w on the same ray and is therefore hidden. (**b**) The self-occlusion test in side view. Vertex v is rejected only if some vertex w (1) projects into the same six-pixel screen cell and (2) is closer to the camera than v by more than 15 mm. Skin adjacent to v lies within 15 mm of it in depth and is not treated as an occluder.

3.2.5. Fixed Hand Shape

Hand shape and hand position trade against each other under a depth objective: a larger hand placed farther away can explain the same measurements as a smaller hand

placed nearer. We therefore fix the shape coefficients to a validated vector and apply them *before* depth-based wrist initialization, holding them frozen through root fitting and smoothing. This is not a post-hoc rescaling of the output mesh. Because the shape enters before the position is estimated, the recovered wrist depth itself changes with it, and a change of shape invalidates the whole optimizer output rather than only its surface.

3.2.6. Temporal Filtering

The optimizer output is preserved unchanged as the auditable result. The trajectory consumed downstream is derived from it by two operations in sequence. First, an isolated translation impulse is repaired: when a sample lies more than 30 mm from the midpoint of its two immediate neighbors while those neighbors remain within 10 mm of each other, it is replaced by that midpoint. Every decision is made from the original sequence, and sustained motion is left untouched by this rule. Second, a tuned One-Euro filter [35] is applied to translation, root rotation, and articulated pose, with filtered rotations projected back to the nearest rotation matrix by singular value decomposition.

3.3. Fingertip-Anchored Retargeting

Retargeting answers one question: where should the robot hand be placed so that it stands in the same relation to the scene as the human hand did. It does not answer what the fingers should do; that comes from teleoperation in Section 3.4. The output of this stage is a six-degree-of-freedom hand root pose per frame, expressed in the simulator camera frame and indexed by the source RGB frame.

The stage is evaluated once per pairing, after the temporal correspondence of Section 3.4. Robot fingertip positions depend on the finger posture, and the posture at each frame is the teleoperation posture that the correspondence assigns to it. A provisional hand-root path, obtained once per episode by retargeting the recovered hand alone, supplies the translation progress used in Section 3.4 and is replaced here.

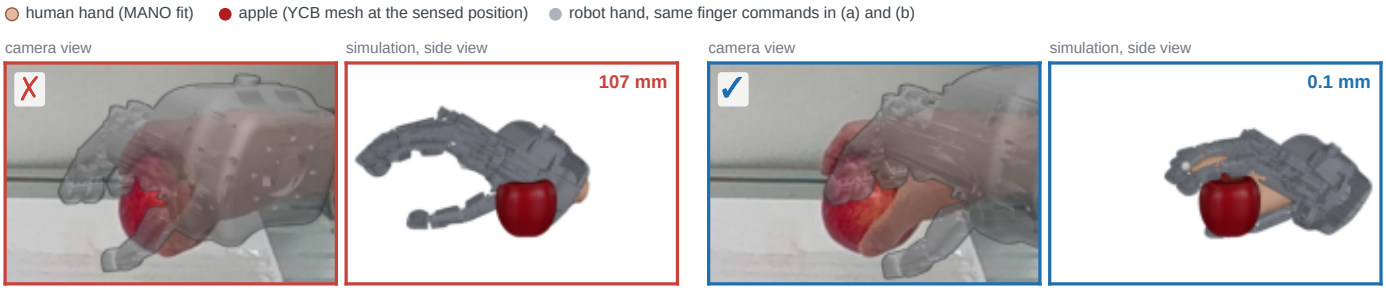
3.3.1. Alignment Target

The parts of a hand that touch an object are its fingertips. We therefore align on the fingertips rather than on the wrist or on the joint angles. Let $p_{1:5}$ be the five human fingertip positions recovered in Section 3.2 and $r_{1:5}$ the corresponding robot fingertip positions in the robot hand's own frame, evaluated by forward kinematics at the finger targets assigned to the same frame. Removing the centroid of each set and solving the orthogonal Procrustes problem [36] yields the rotation that best carries the robot tips onto the human tips, and the centroid difference yields the translation:

$$R^\star = \arg\min_{R \in SO(3)} \sum_{i=1}^{5} \left\| R(r_i - \bar{r}) - (p_i - \bar{p}) \right\|^2, \tag{4}$$

$$t^\star = \bar{p} - R^\star \bar{r}. \tag{5}$$

Because the alignment is computed on the contact points themselves, a robot hand whose proportions differ from the human hand approximately preserves the human's grasp location, which is the property the downstream task depends on. We refer to this as fingertip-anchored retargeting. Figure 5 contrasts it with placing the robot wrist at the human wrist: with identical finger commands, wrist placement leaves the grasp far from where the human grasped, whereas fingertip anchoring moves the wrist back and keeps the grasp in place.



(a) Wrist-aligned placement **(b) Fingertip-anchored placement (ours)**

Figure 5. Why the hand root is anchored at the fingertips. The robot hand replays the same recorded finger commands in (**a**) and (**b**); only the placement of the hand root differs. The two hands differ in size: the wrist-to-middle-fingertip length is 146 mm for the demonstrator and 268 mm for the robot. (**a**) Wrist-aligned placement: with the robot wrist placed at the human wrist, the robot fingertips overshoot the object. (**b**) Fingertip-anchored placement: with the five fingertips anchored, the wrist retreats by the length the robot hand requires and the grasp stays on the object. The printed values are the grasp-center offset at the frame shown, the distance between the centroid of the five robot fingertips and that of the five human fingertips; over the whole demonstration it averages 109 mm in (**a**) and 0.8 mm in (**b**). Fingertip anchoring registers the two centroids, so the offset in (**b**) is small by construction; the values quantify what is lost without anchoring rather than the accuracy of the anchoring itself. The apple mesh is drawn only to visualize the scene; the pipeline takes no object mesh or object pose as input.

3.3.2. Wrist-Direction Correction

Fingertip alignment fixes where the contact points of the robot hand are, but it does not constrain the direction from which the forearm reaches them: two hands of different proportions can agree at the tips while their wrists point in visibly different directions. One additional rotation therefore brings the robot wrist direction toward the human one. Let w_h be the unit vector from the human fingertip centroid to the human wrist, and w_r the unit vector from the robot fingertip centroid to the robot wrist after the rotation $R^\star$. Let ν be the correction axis, taken as the right singular vector (in the camera frame) with the smallest singular value of the fingertip cross-covariance $\sum_i (r_i - \bar{r})(p_i - \bar{p})^\top$. Both wrist directions are projected onto the plane orthogonal to ν, $w^\perp = w - (\nu^\top w)\,\nu$, and the correction is the rotation about ν by the signed angle between the projections:

$$\theta = \operatorname{atan2}\big(\nu^\top (w_r^\perp \times w_h^\perp),\ (w_r^\perp)^\top w_h^\perp\big), \qquad R = \operatorname{Rot}(\nu, \theta)\, R^\star. \tag{6}$$

This step is a heuristic. A rotation about ν moves the fingertips, so it gives up some fingertip agreement in exchange for a forearm direction consistent with the video, and we do not claim that it is the optimal trade-off between the two. The resulting rotations are smoothed over a short window, the translation is recomputed from Eq. (5) with the smoothed rotation so that the fingertip centroids still coincide, and the hand-root trajectory then receives the global fit of Appendix A.

3.3.3. Separation of Hand-Root Retargeting and Finger Commands

This stage estimates the hand-root trajectory without an object pose or object mesh. Finger motion is supplied separately by the recorded teleoperation commands. Robot-specific kinematics map those native commands into the simulator representation; the temporal correspondence of Section 3.4 and the output scheduling of Section 3.5 evaluate the same command path at new times.

3.4. PHTA: Persistent Hand-State Transition Alignment

The human episode and the teleoperation demonstration perform the same task at different speeds, with different pauses, and with different amounts of time spent in each phase. Before the two can be composed, one must know which moment in the video corresponds to which moment in the log.

Human and robot hands differ in joint count, link lengths, and joint axes. Direct joint-angle matching is therefore inappropriate, and absolute workspace coordinates need not coincide across independently recorded trials. PHTA instead represents both hands in a common dimensionless shape space (Section 3.4.1 and Figure 6), associates ordered multi-scale transitions using deformation and trajectory-context evidence, and aligns progress within the matched transitions. The resulting correspondence transfers the recorded robot finger commands to the human episode (Figure 7).

Here, *source* denotes the human recording and *target* denotes the teleoperation recording supplying the finger commands.

3.4.1. A Common Intrinsic Hand State

For each frame and finger, we use three ordered distal-chain landmarks in a palm-local frame: the start and intermediate point of the distal chain, and the fingertip; on the human hand these are the PIP and DIP joints and the fingertip for the index to little fingers, and the MCP and IP joints and the fingertip for the thumb. Human landmarks are obtained from the reconstructed hand model. Robot landmarks are evaluated by forward kinematics at the recorded finger position targets. The same three-landmark convention is used for each of the five fingers, with robot-specific landmark definitions supplying the corresponding geometric measurements.

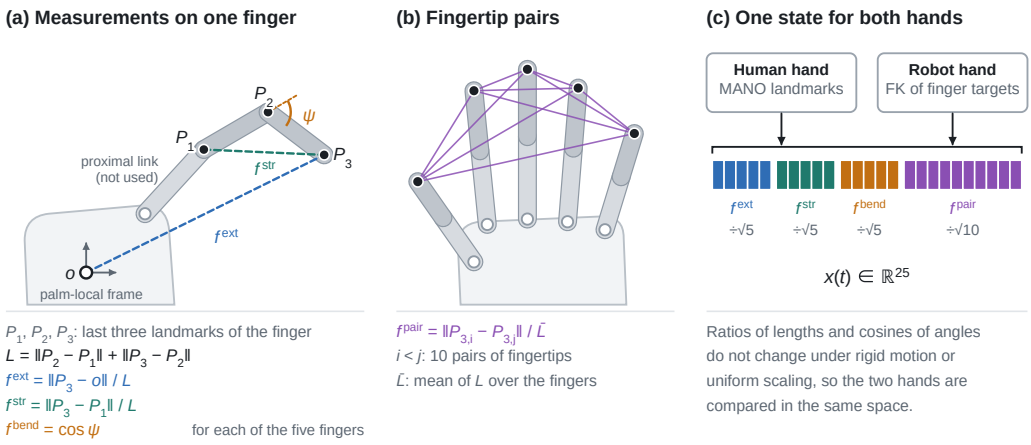


Figure 6. Intrinsic hand state of PHTA. (**a**) On each finger, the last three landmarks P_1, P_2, P_3 in the palm-local frame define the two distal links; the palm-origin-to-tip distance (f^{ext}) and the distance from P_1 to the tip (f^{str}) are divided by the chain length L, and f^{bend} is the cosine of the angle ψ between the two links. (**b**) The ten fingertip-pair distances (f^{pair}) are divided by the mean chain length $\bar{L}$. (**c**) Human landmarks from the hand model and robot landmarks from forward kinematics (FK) at the recorded finger targets are mapped into the same 25-dimensional vector $x(t)$ of Eq. (7), each group divided by the square root of its size.

Four dimensionless feature groups are computed from these landmarks at each frame: per finger, the palm-origin-to-tip distance divided by the summed length of the two distal links; per finger, the straight-line distance from the start of the distal chain to the tip, divided by the same sum; per finger, the cosine of the angle between consecutive link

directions; and per fingertip pair, their distance divided by the mean distal chain length (Figure 6 and Table 3). Writing these as f^{ext}, f^{str}, f^{bend} and f^{pair},

$$x(t) = \left[\frac{f^{\text{ext}}(t)}{\sqrt{5}}, \frac{f^{\text{str}}(t)}{\sqrt{5}}, \frac{f^{\text{bend}}(t)}{\sqrt{5}}, \frac{f^{\text{pair}}(t)}{\sqrt{10}} \right] \in \mathbb{R}^{25}. \tag{7}$$

Each group is divided by the square root of its dimension so that pairwise distances do not receive greater weight solely because they have more coordinates. The descriptor is invariant to rigid hand pose and uniform spatial scaling. Differences in finger proportions and joint structure remain; PHTA therefore uses transition evidence rather than requiring equal human and robot descriptor values at corresponding times.

Table 3. Components of the intrinsic hand state $x(t)$.

Group	Definition	Dim.
f^{ext}	palm-origin-to-tip distance / summed length of the two distal links	5
f^{str}	distal-chain start-to-tip distance / the same sum	5
f^{bend}	cosine of the angle between consecutive link directions	5
f^{pair}	fingertip-pair distance / mean distal-chain length	10
Total		25

3.4.2. Events That Persist Across Time Scales

Each episode's state trajectory is resampled onto a normalized phase grid of 257 samples. For a half-width h and a grid center n_0, the transition deformation is the difference between the mean state after and before the center:

$$\Delta_h(n_0) = \frac{1}{h} \sum_{n=n_0+1}^{n_0+h} x_n - \frac{1}{h} \sum_{n=n_0-h}^{n_0-1} x_n. \tag{8}$$

This deformation is normalized by the L_2 norm of the episode's per-feature range. Local maxima of its norm are observed at six half-widths, $\{2, 4, 8, 16, 32, 64\}$, and observations at adjacent scales are linked into a branch. At a merge, the branch with the largest accumulated squared deformation inherits the coarser observation, while the other branches remain in the candidate inventory. This accumulated-energy rule is inspired by multiscale persistence [26,27]. Equation (9) defines the energy E_b and unit deformation direction v_b from the scale observations ℓ of branch b.

$$E_b = \sum_{\ell \in b} \|\Delta_{b,\ell}\|_2^2, \qquad v_b = \frac{\sum_{\ell \in b} \Delta_{b,\ell}}{\left\| \sum_{\ell \in b} \Delta_{b,\ell} \right\|_2}. \tag{9}$$

3.4.3. How Many Events, and Which Ones

Calibration pools branch log-energies across recordings separately for human and robot hands. A Gaussian mixture is fitted to each population, with its component count selected by BIC [28], and held fixed during pairwise alignment.

Target branches assigned by maximum posterior to the component with the largest mean form the target inventory $\mathcal{B}^T_{\text{persistent}}$, whose size K is given by Eq. (10). All source branches remain candidates.

$$K = \left| \mathcal{B}^T_{\text{persistent}} \right|. \tag{10}$$

Source branch i contributes its posterior ρ_i of membership in the highest-mean source component. For that component's index $k^\star$ and mixture weights, means, and variances ω_k, μ_k, and σ_k^2, this posterior is computed from the branch log-energy as follows:

$$\rho_i = \frac{\omega_{k^\star}\,\mathcal{N}(\log E_i;\, \mu_{k^\star}, \sigma_{k^\star}^2)}{\sum_k \omega_k\,\mathcal{N}(\log E_i;\, \mu_k, \sigma_k^2)}. \tag{11}$$

For source and target unit deformation directions v_i^S and v_j^T, directional agreement is scored by a von Mises–Fisher density f_{vMF} centered on v_j^T relative to the uniform density f_{unif} on the unit sphere in $\mathbb{R}^{25}$ [29]. Its log-density ratio is $\kappa\,(v_i^S)^\top v_j^T + c_{25}(\kappa)$, where the concentration κ controls directional selectivity and c_{25} follows from the density normalizers.

Calibration fits an unsupervised mixture of aligned and uniform directions using candidate direction pairs from high-energy source and target branches within the supplied episode pairings. The fitted κ is held fixed during alignment.

3.4.4. Trajectory Context

Shape evidence alone can be ambiguous, because similar hand configurations recur within one task. A third statistical term places each event in the trajectory rather than only in shape space.

For each episode, cumulative hand-root translation length is divided by total path length to give progress $z \in [0, 1]$. Source translation comes from the provisional per-episode hand-root path (Section 3.3); target translation comes from forward kinematics of the recorded arm joints. This fraction of distance traveled is invariant to rigid transforms and uniform scale and retains traversal order.

Each branch's multiscale time supports map to intervals on z. Uniform densities on these intervals are weighted by each scale's contribution to branch energy, giving normalized densities $g_i^S(z)$ and $g_j^T(z)$ for source branch i and target branch j. Their overlap supplies the context term:

$$\Gamma_{ij} = \int_0^1 g_i^S(z)\, g_j^T(z)\, dz. \tag{12}$$

The overlap Γ_{ij} is large when both events concentrate on the same region of normalized translation progress. It is a density-overlap score and may exceed one; its logarithm contributes to the association score.

3.4.5. Assignment

The composite association score adds source-event log-odds, the directional log-density ratio, and log translation-progress overlap:

$$S_{ij} = \log\frac{\rho_i}{1-\rho_i} + \log\frac{f_{\text{vMF}}(v_i^S \mid v_j^T, \kappa)}{f_{\text{unif}}(v_i^S)} + \log\Gamma_{ij}. \tag{13}$$

The K target events, in time order, are assigned an increasing sequence of source branches maximizing the total score:

$$(i_1^\star, \ldots, i_K^\star) = \underset{i_1<\cdots<i_K}{\arg\max} \sum_{j=1}^{K} S_{i_j j}. \tag{14}$$

Dynamic programming solves Eq. (14) exactly under the order constraint, assigning one distinct source branch to each selected target event. Every source branch is eligible; all K target events receive a match.

3.4.6. Continuous Time Inside an Event

Midpoints between consecutive matched event centers define corresponding cells, with the episode endpoints delimiting the first and last cells. The centers determine this partition; cumulative measures within each cell determine the temporal correspondence. These measures start from scalar projections of the descriptor onto the matched branch directions, for source and target sample indices n and m:

$$a_n^S = (x_n^S)^\top v^S, \qquad a_m^T = (x_m^T)^\top v^T. \tag{15}$$

Isotonic regression [30] fits nondecreasing scalar signals $\hat{a}^S$ and $\hat{a}^T$ to these projections by least squares for timing estimation. On the source side, squared positive increments give energy e_n^S, weighted by γ_n to obtain timing mass w_n^S:

$$e_n^S = \left[\max(\hat{a}_{n+1}^S - \hat{a}_n^S, 0)\right]^2, \qquad w_n^S = e_n^S \gamma_n. \tag{16}$$

Within each source cell, a Gaussian component plus a uniform background is fitted to normalized increment times $\xi_n \in [0,1]$, using e_n^S as observation weights. The responsibility γ_n of the Gaussian component is given below, with mixture weight η, mean μ, variance σ^2, and unit background density.

$$\gamma_n = \frac{\eta\,\mathcal{N}(\xi_n;\,\mu,\sigma^2)}{\eta\,\mathcal{N}(\xi_n;\,\mu,\sigma^2) + (1-\eta)}. \tag{17}$$

Squaring and localization emphasize concentrated changes to reduce the influence of distributed reconstruction drift. Target timing mass instead follows the positive increments of the fitted target signal:

$$w_m^T = \max(\hat{a}_{m+1}^T - \hat{a}_m^T, 0). \tag{18}$$

The source mass localizes the transition in time, while target increments describe progress along the command-induced hand-shape path. Let $\Lambda^S(t)$ and $\Lambda^T(\tau)$ be their normalized cumulative sums at source and teleoperation times t and τ. Matching equal quantiles s aligns these two cumulative measures:

$$\Lambda^S(t) = \Lambda^T(\tau) = s, \qquad s \in [0,1]. \tag{19}$$

The cumulative-measure breakpoints and original target command samples are retained as correspondence knots. Writing the correspondence as $t = \varphi(\tau)$, the complete recorded command vector $u_T(\tau)$ is placed at source time t. All finger channels share this map and retain their recorded command path. Interpolation and velocity-constrained scheduling produce the final fixed-rate output (Section 3.5). Figure 7 shows the resulting correspondence through geometric traces of the human and robot hands.

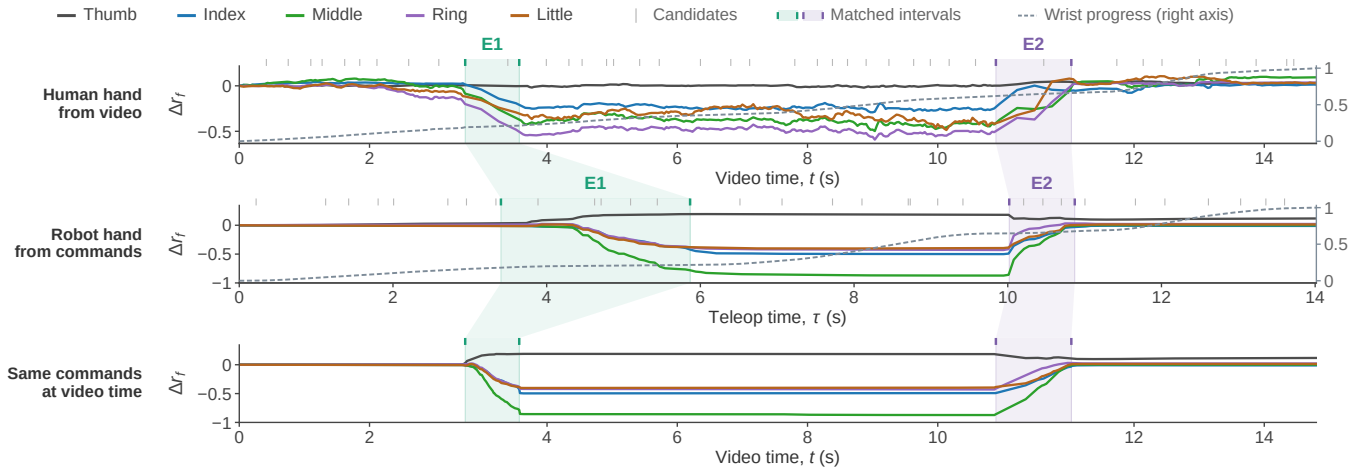


Figure 7. Persistent hand-state transition alignment for a human–teleoperation pair. Top: hand state recovered from video. Middle: robot hand state evaluated from recorded finger targets. Bottom: the same robot-state trajectories re-timed on the human recording axis, before output scheduling. The five solid curves show Δr_f: the change from the first frame in palm-to-tip distance normalized by distal-chain length, one curve per finger. Gray ticks mark all candidate centers. Green and purple spans show the 5th–95th percentiles of the respective timing masses for matched events E1 and E2; ribbons connect these display intervals. Computational cell boundaries are omitted. Dashed curves show normalized cumulative wrist-path length on the right axes. The highlighted intervals on the top and bottom time axes are expanded eightfold, with tick labels retaining the original times; widths and slopes therefore use a nonuniform display scale.

3.4.7. Inputs and Assumptions

Given compatible source–teleoperation episodes, PHTA takes human hand geometry and hand-root translation, recorded robot finger targets and arm-joint states, and the fixed calibration. It requires no annotated task phases, grasp or release times, or contact labels. The same robot-joint input interface supports virtual-reality and leader-arm teleoperation.

3.5. Velocity-Constrained Output Scheduling

The correspondence defines a path in source and teleoperation coordinates. The source hand-root path is smoothed before inverse kinematics; its native samples are merged with the dense correspondence knots and original finger-command samples.

For interval i of the merged path, let Δt_i^S be the elapsed source time. The changes in arm joint j, Δq_{ij}^A, and finger target j, Δu_{ij}^H, divided by their velocity limits $\dot{q}_j^{A,\max}$ and $\dot{u}_j^{H,\max}$, give the required traversal durations:

$$\Delta t_i^A = \max_j \frac{|\Delta q_{ij}^A|}{\dot{q}_j^{A,\max}}, \qquad \Delta t_i^H = \max_j \frac{|\Delta u_{ij}^H|}{\dot{u}_j^{H,\max}}. \tag{20}$$

The interval receives the longest of the three durations:

$$\Delta t_i = \max\left(\Delta t_i^S,\ \Delta t_i^A,\ \Delta t_i^H\right). \tag{21}$$

Durations are accumulated before sampling once at the output rate. The reported configuration uses source and output rates of 30 and 15 Hz, and arm and finger limits of 20 and 90 degrees/s. Root smoothing, support merging, and output checks are detailed in Appendix A.

3.6. Part-Aware Visibility Transfer and Compositing

The last stage produces the image. The human arm and hand are removed from the source frame and the retargeted robot arm and hand are drawn in their place. The difficulty is not the drawing but the occlusion: when the hand and the manipulated object overlap in the image, some parts of the hand are in front of the object and some are behind it, and a robot hand pasted without that distinction reads as a sticker laid over the scene.

The usual way to resolve occlusion is to compare depths of known three-dimensional geometry. That is unavailable here by design: we do not assume an object mesh, and we do not estimate a six-degree-of-freedom object pose. What we do have is the original human demonstration, in which the correct occlusion relation was recorded by the camera. The stage therefore reads that relation from the human video and transfers it to the robot hand.

3.6.1. Three Kinds of Evidence, None Substitutable

Let Ω be the image domain and $\mathcal{P}_H$, $\mathcal{P}_R$ the semantic part sets of the human and robot hands. At time t the stage consumes three masks: $H_{t,p} \subseteq \Omega$, the projection of MANO part p into the image, rendered with depth ordering so that each pixel carries the hand part nearest to the camera, which says where each hand part lies and which part an unobserved region would have belonged to; $M_t^H \subseteq \Omega$, the human hand actually observed in the frame, which says what was visible but not which part a hidden region belongs to; and $O_t \subseteq \Omega$, the object's amodal silhouette including the portion the hand covers, which delimits where a front-or-behind decision is needed at all.

None of the three replaces another. The projection supplies part identity and continuity, the visible mask supplies observed visibility, and the silhouette supplies the region in which occlusion by the object is possible. Only in combination do they answer the question the stage must answer: which hand part overlaps the object, and which portion of it was actually seen.

3.6.2. Reading the Relation from the Human Video

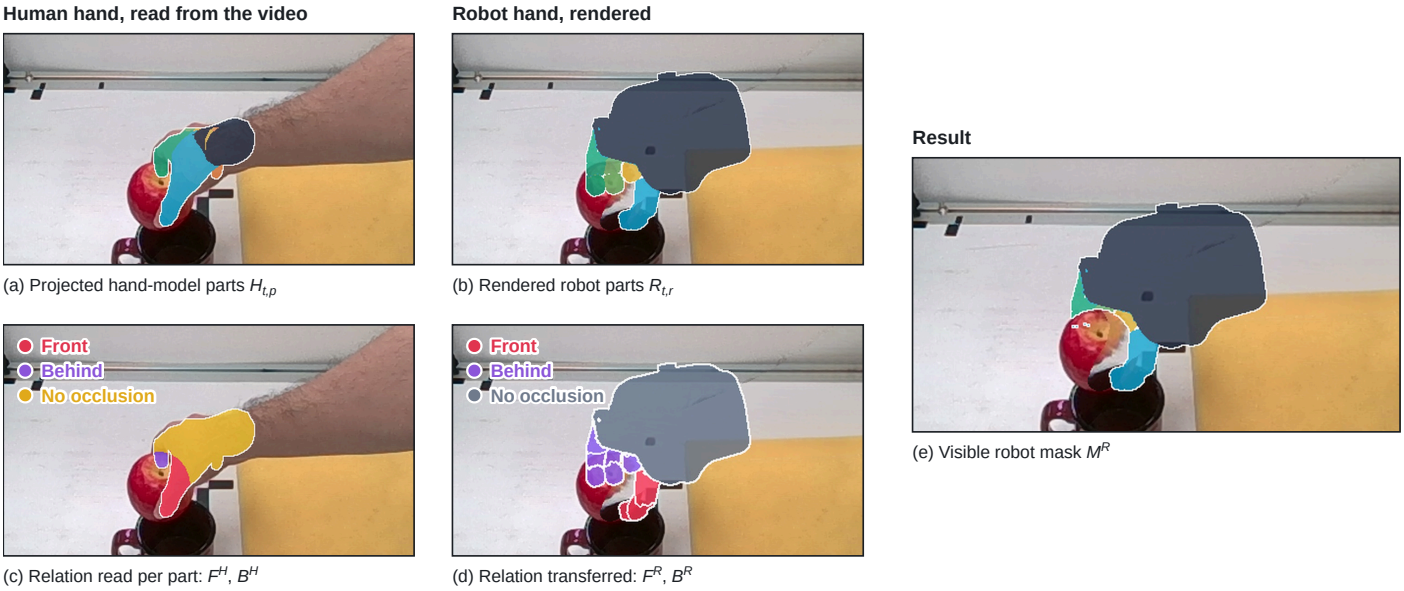
A hand region outside the object needs no decision, so inference is restricted to the interaction region

$$X_{t,p}^H = H_{t,p} \cap O_t, \tag{22}$$

which the observed hand splits into two disjoint parts,

$$F_{t,p}^H = X_{t,p}^H \cap M_t^H, \qquad B_{t,p}^H = X_{t,p}^H \setminus M_t^H, \tag{23}$$

so that $X_{t,p}^H = F_{t,p}^H \dot{\cup} B_{t,p}^H$. A region that overlaps the object and was nevertheless observed is evidence that the part is in front of the object. A region that the hand model places inside the object silhouette but the camera did not see is evidence that the part is behind it. The formulation deliberately does not treat every unobserved hand region as occluded by the object: occlusion is defined only where the part projection and the object silhouette overlap, so a part hidden by the hand itself, or lying outside the frame, does not enter the decision. Figure 8 summarizes the decision.



(a) Projected hand-model parts $H_{t,p}$

(b) Rendered robot parts $R_{t,r}$

(e) Visible robot mask M^R

(c) Relation read per part: F^H, B^H

(d) Relation transferred: F^R, B^R

Figure 8. Part-aware visibility transfer on one frame. (**a**) The projected hand-model parts $H_{t,p}$ and (**b**) the rendered robot parts $R_{t,r}$, colored by part. (**c**) The relation read from the human video: inside the object silhouette O_t, a part region that was observed is in front of the object (F^H) and a region that the hand model places there but the camera did not see is behind it (B^H); parts that do not overlap the object need no decision. (**d**) The same relation transferred to the corresponding robot parts through the part correspondence χ (F^R, B^R). (**e**) The visible robot mask M_t^R of Eq. (25), from which the robot regions behind the object are removed. Panels (b), (d) and (e) are drawn on the background from which the human has been removed. No object mesh or object pose is used.

3.6.3. Transferring the Relation, Not the Pixels

Human and robot hands have different shapes, so the pixels of F^H and B^H cannot be copied into the robot image. What transfers is the relation, carried across a semantic part correspondence $\chi : \mathcal{P}_H \to \mathcal{P}_R$, so that a conclusion drawn about, say, the proximal link of the human index finger is applied to the corresponding link of the robot index finger (Figure 9). Writing $R_{t,r}$ for the rendered robot part mask, the transfer operator $\mathcal{T}_p$ converts the human partition into a partition of the robot part's interaction region,

$$\left(F^R_{t,\chi(p)},\, B^R_{t,\chi(p)}\right) = \mathcal{T}_p\left(F^H_{t,p},\, B^H_{t,p};\; R_{t,\chi(p)},\, O_t\right), \tag{24}$$

with $F^R_{t,\chi(p)} \,\dot\cup\, B^R_{t,\chi(p)} = R_{t,\chi(p)} \cap O_t$. The operator decides each robot link from the evidence of its human counterpart. The human part is taken to be in front of the object when a sufficient portion of its interaction region was observed and only a small portion was not. In that case the robot front region $F^R_{t,\chi(p)}$ is the part of the robot interaction region that is connected to pixels where the human part was observed; for the palm it is the whole interaction region. Otherwise the robot link is placed behind the object inside the silhouette. For the palm and the proximal finger links the evidence is pooled with neighbouring parts, which stabilizes regions that are only partly visible. The roles are thus separated: the human video supplies the hand–object visibility relation, the robot rendering supplies the final shape and pixel positions, and the part correspondence carries the relation between the two representations.

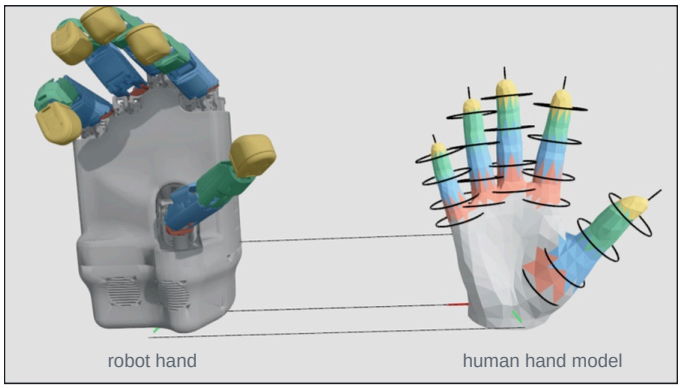


Figure 9. Common part convention for the robot hand and the human hand model. Equal colors mark corresponding finger segments, from the proximal segment to the fingertip. The two hands differ in size, proportions, and joint structure; the part correspondence χ maps each human part to the robot part of the same color, and the front-or-behind relation read from a human part is applied to its robot counterpart.

3.6.4. Compositing

The robot hand is visible where it does not overlap the object, and, where it does, only where the transferred relation says front:

$$M_t^R = \bigcup_{r \in \mathcal{P}_R} \Big[(R_{t,r} \setminus O_t) \cup F_{t,r}^R \Big]. \tag{25}$$

With C_t the clean plate from which the human has been removed, G_t the rendered robot image and α_t^G its alpha, the composite is

$$\alpha_t^\star(x) = \alpha_t^G(x)\, \mathbf{1}\big[x \in M_t^R\big], \tag{26}$$

$$\tilde{I}_t(x) = \alpha_t^\star(x)\, G_t(x) + \big(1 - \alpha_t^\star(x)\big)\, C_t(x). \tag{27}$$

Where the relation says behind, the robot alpha is removed and the object in the clean plate is revealed; where it says front, the render is kept. The human hand's appearance is gone, but the spatial relation it had with the object survives in the robot hand. The whole stage can be read as one chain,

$$(H_{t,p}, M_t^H, O_t) \to (F^H, B^H) \xrightarrow{\mathcal{T}_p} (F^R, B^R) \to \tilde{I}_t. \tag{28}$$

3.6.5. Why Not Pixel Matching

An apparently simpler rule would ask, for each rendered robot pixel, whether the MANO part map carries the same part at that pixel. We reject that rule. MANO and the retargeted robot geometry are not pixel-aligned—they are different shapes placed by a retargeting that matches contact points, not silhouettes—so such a test measures shape disagreement rather than occlusion. The distinction matters when diagnosing an output: the rendered robot pixels are where a decision is *applied*, never evidence *for* it. The evidence is only the MANO part map, the visible human-hand mask, and the object silhouette.

3.6.6. Producing the Masks

The visible human-hand mask is the intersection of a promptable segmentation model [22] with the MANO projection, which suppresses segmentation bleed onto the forearm and background while keeping the region the hand model accounts for. The amodal object silhouette is obtained in two steps: a text prompt yields the visible object mask through the promptable segmentation model [22], and a video amodal segmentation

model [23] completes it into the full silhouette, including the portion covered by the hand. Neither step requires an object mesh or an object pose; a mesh-projection backend [25] remains available for objects with a registered model, but it is a compatibility path and not the configuration reported here. The clean plate is produced by video inpainting [24] over the human arm and hand.

Because inpainting depends only on the source frames and the human mask, it is solved once per human episode and reused by every teleoperation demonstration paired with it, each selecting the cached frames through its own source-frame indices.

This is where the mesh-free property of the title is earned. Occlusion is decided from part labels and masks rather than from the depth ordering of known three-dimensional geometry, so a new object needs a text prompt rather than a scanned model, and nothing in the pipeline requires the object's pose to be estimated at all.

3.7. Experimental Setup

The robot is a Doosan Robotics M0609 six-axis arm with an AIDIN ROBOTICS Gen1 five-fingered hand with 15 degrees of freedom mounted at the wrist, used on the right side. Human episodes are recorded with an Intel RealSense D455 RGB-D camera at 1280×720 pixels and stored as an 848×480 crop of the workspace region, with the depth stream registered to the color pixels. The same camera stays fixed in place for recording the human episodes and for evaluating the policies on the robot. Teleoperation is recorded on the same robot in separate sessions, with a leader arm for the sweeping and pouring tasks and with HTC Vive trackers for the apple and cylinder tasks. The source runs at 30 Hz and the composed output at 15 Hz, with a $20^\circ/\text{s}$ arm limit and a $90^\circ/\text{s}$ hand limit.

The robot arm and hand are rendered in a Unity simulator from the meshes of the robot's Unified Robot Description Format (URDF) model. The simulator camera uses the intrinsics of the real camera and the camera–robot calibration, so the rendered robot appears where the real robot would appear in the camera image.

Four tasks are evaluated: placing an apple in a cup, placing a cylinder in a box, sweeping with a brush, and pouring from a cup (Figure 10). For each task, 20 human episodes and 5 teleoperation demonstrations are composed into 100 demonstrations. Every data condition trains the same visuomotor diffusion policy [37], with a flow-matching objective [38], for 1000 epochs. The policy observes the camera image and the 21-dimensional joint state (six arm joints and fifteen finger targets) over two frames, predicts the next 16 joint targets, and executes the first 8. Each policy is evaluated in 20 trials per task on the physical robot, with the objects placed at nearly the same initial position in every trial and the same initial positions used for all data conditions.

3.8. Evaluation Protocol

3.8.1. Main Comparison

The central question is whether composing human video with a small amount of teleoperation produces better policies than that teleoperation alone. The comparison is therefore controlled on the *teleoperation budget*: the composed condition uses the same five teleoperation demonstrations as the teleoperation-only baseline, and additionally human video, which is the inexpensive input.

- *Teleop-only.* Policies trained on the five teleoperation demonstrations as recorded. This is the conventional pipeline at the same teleoperation budget.
- *Human-only.* Policies trained on 100 recorded human episodes in which the finger commands are derived from the human hand reconstruction, by retargeting the fingertip positions through inverse kinematics, rather than taken from teleoperation, with the rest of the pipeline unchanged. This isolates the executability gap: it measures

what is lost when the composition of Section 3.4 is removed and the video is asked to supply the whole robot state.

- *Ours.* Policies trained on the 20×5 composed demonstrations.

As a reference, policies are also trained on 100 teleoperation demonstrations, twenty times the teleoperation budget of the other conditions. Apart from Teleop-only, every condition therefore trains on 100 demonstrations; Ours uses 20 human episodes so that, paired with the five teleoperation demonstrations, it reaches the same number.

3.8.2. Ablations

Three components are removed in turn while the rest of the pipeline is held fixed. *Depth supervision*: the masked dense term of Section 3.2 is disabled, and the recovered hand trajectory is compared with the measured depth surface. *Temporal alignment*: PHTA is compared with pairwise dynamic time warping [31] over the same episode pairs. *Compositing*: part-aware visibility transfer is replaced by pasting the robot render without occlusion reasoning, and by inpainting the human without drawing a robot.

3.8.3. Alignment Evaluation

The alignment evaluation is separate from the behavior cloning experiments and uses its own recordings of two tasks (Section 4.3). Alignment is scored against four hand boundaries annotated manually on the original RGB of every human and teleoperation recording: the start and the completion of hand closing, and the start and the completion of hand opening. Each annotation is an interval that reflects its visual uncertainty. A boundary of a teleoperation recording is mapped through the estimated correspondence to the human timeline, and its error is the absolute difference in seconds from the annotated human boundary. Boundaries that cannot be read reliably are excluded for both methods alike. Besides the median, 90th percentile, and maximum error, a method is counted as better on a boundary only when its largest possible error under the annotation intervals is smaller than the smallest possible error of the other method; the remaining boundaries are counted as unresolved. The evaluation uses the frozen calibration without any parameter change.

The baseline is exact pairwise dynamic time warping [31] on the same 25-dimensional hand state, with a Euclidean local cost, a full monotone lattice without a band constraint, and fixed endpoints; the path is converted to a function by averaging the source indices assigned to each target sample and interpolating linearly. Both methods feed the same output scheduling.

4. Results

4.1. Main Comparison

Table 4 reports the success rate of each data condition, and Figure 10 shows composed frames for each task. Five teleoperation demonstrations alone are too few to cover the task: the resulting policies succeed in 20% and 45% of the trials on the apple and cylinder tasks and never on sweeping and pouring. Composed with 20 human episodes, the same five demonstrations reach 80%, 90%, 55%, and 50%. On the apple and cylinder tasks this equals the policy trained on 100 teleoperation demonstrations, and on sweeping it is within one trial of it; the largest remaining gap is on pouring, 50% against 70%. The human-only condition, which derives the finger commands from the human hand, reaches only 5–60% and is below the composed data on every task, although it uses five times as many human episodes. By Fisher's exact test, the improvement over Teleop-only is significant on all four tasks ($p < 0.01$), and no task shows a significant difference from the 100-demonstration reference ($p \geq 0.33$). Mixing human video into five teleoperation demonstrations therefore

attains the same observed success rate as one hundred teleoperation demonstrations on two tasks.

Table 4. Main comparison. Success rate of policies trained under each data condition, given as successes/trials (%). The first three rows use at most five teleoperation demonstrations; the last row is a reference with twenty times the teleoperation budget.

	Training data (count)			Task			
Condition	Teleop.	Human	Demos	Apple in cup	Cylinder in box	Sweeping	Cup pouring
Teleop-only	5	0	5	4/20 (20)	9/20 (45)	0/20 (0)	0/20 (0)
Human-only	0	100	100	6/20 (30)	12/20 (60)	1/20 (5)	3/20 (15)
Ours	5	20	100	**16/20 (80)**	**18/20 (90)**	**11/20 (55)**	**10/20 (50)**
Teleop-only, 20× budget	100	0	100	16/20 (80)	18/20 (90)	12/20 (60)	14/20 (70)

Teleop., Human and Demos give the numbers of teleoperation demonstrations, human episodes and training demonstrations. Human-only derives the finger commands from the human hand reconstruction instead of teleoperation. Ours composes 20 human episodes with 5 teleoperation demonstrations into 100 demonstrations. Bold marks the best result among the first three rows.

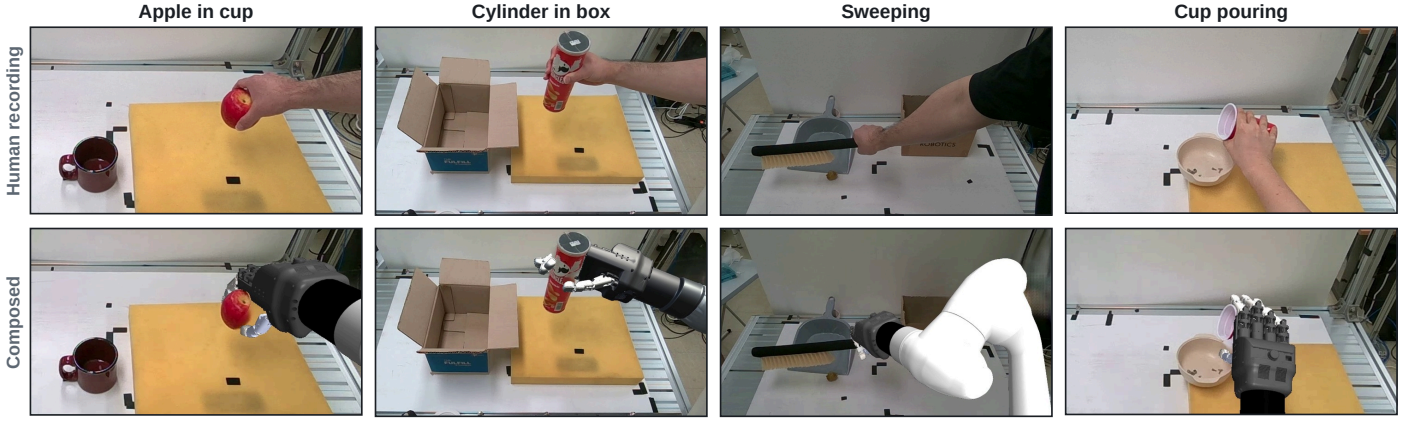


Figure 10. Composed demonstrations for four tasks: placing an apple in a cup, placing a cylinder in a box, sweeping with a brush, and pouring from a cup. Top: a frame of the human RGB-D recording. Bottom: the corresponding composed frame, in which the human hand and arm are removed and the robot arm and hand are drawn at the anchored hand-root pose with the transferred visibility relation. Both frames of a column use the same crop.

4.2. Ablations

Table 5 reports the depth supervision ablation and Table 6 the compositing ablation; the temporal alignment comparison is given in Section 4.3. Depth supervision lowers the median surface residual from 15.3 to 11.1 mm. In the compositing ablation, pasting the robot render without occlusion reasoning lowers the success rate from 80% to 50% on the apple task and from 90% to 75% on the cylinder task, and removing the human without drawing a robot lowers it further, to 20% and 55%.

Table 5. Effect of depth supervision on the recovered hand trajectory: surface residual between the recovered hand and the measured depth, in mm.

Configuration	Median	95th pct.	Max
No depth term (image evidence only)	15.3	29.9	40.6
Masked dense depth (reported)	**11.1**	**21.6**	**32.9**

The surface residual is the median over the gated vertices of each frame (about 150 per frame) of the absolute difference between predicted and measured depth. One apple-in-cup episode, 191 frames sampled every third frame of 572.

Table 6. Compositing ablation. Success rate of policies trained on data composed with each variant, given as successes/trials (%).

Variant	Apple in cup	Cylinder in box
Part-aware visibility transfer (full)	16/20 (80)	18/20 (90)
Robot render pasted without occlusion reasoning	10/20 (50)	15/20 (75)
Human removed, no robot drawn	4/20 (20)	11/20 (55)

A separate diagnostic concerns trajectory quality rather than a design choice. On a recorded episode the raw optimizer output reached a maximum inter-frame translation of 162.3 mm; the isolated-impulse repair of Section 3.2 selected exactly two samples, reducing the maximum to 25.0 mm before the One-Euro filter and 10.8 mm after it, with the trajectory endpoints unchanged.

4.3. Temporal Alignment Accuracy

Table 7 reports the alignment of two tasks: placing an apple in a cup, with 21 human and 7 teleoperation recordings (147 pairs), and pouring from a cup, with 25 human and 5 teleoperation recordings (125 pairs). On the apple task PHTA places the hand boundaries with a median error of 0.06 s, about two frames, against 2.03 s for dynamic time warping, and its largest error, 0.52 s, is smaller than the median error of the baseline. Under the annotation intervals PHTA is better on 515 of 546 boundaries and the baseline on none. On the pouring task the median errors are 0.37 and 0.90 s and the 90th-percentile errors 1.36 and 7.17 s: PHTA mainly removes the large misalignments of the baseline, and is better on 254 boundaries against 87. The gain is not uniform. At the completion of hand opening in the pouring task the baseline is more accurate (median 0.20 against 0.45 s, maximum 1.30 against 6.69 s), and the largest PHTA error in that task is 10.49 s, at the completion of a closing. Appendix B gives the errors for each of the four boundaries (Table A1).

The output schedule preserves the estimated correspondence: over 83,162 output samples the continuous source coordinate differs from the correspondence by at most 0.003 source frames.

Table 7. Temporal alignment of hand boundaries. Errors are absolute differences, in seconds on the human timeline, between the mapped teleoperation boundary and the manually annotated human boundary; lower is better. The last three columns count the boundaries on which one method is better under the annotation intervals, and those that remain unresolved.

			Error (s)			Better on boundaries		
Task	Method	Boundaries	Median	90th pct.	Max	PHTA	DTW	Unresolved
Apple in cup	PHTA (ours)	546/588	**0.061**	**0.192**	**0.517**	515	0	31
	Pairwise DTW	546/588	2.033	3.817	5.233			
Cup pouring	PHTA (ours)	405/500	**0.372**	**1.358**	10.491	254	87	64
	Pairwise DTW	405/500	0.900	7.167	**10.233**			

Apple in cup: 21 human and 7 teleoperation recordings, 147 pairs. Cup pouring: 25 human and 5 teleoperation recordings, 125 pairs. Each pair contributes four boundaries (start and completion of closing and of opening); Boundaries gives those that could be read reliably over all, and the same boundaries are excluded for both methods. Bold marks the smaller value within a task.

4.4. Cost

The reuse boundary of Section 3.1 means that object masks, visible hand masks, hand trajectory recovery, the provisional hand-root path, and human removal are computed N times to produce $N \times M$ demonstrations. Table 8 reports the wall-clock time of each stage for one sweeping episode of 19.1 s (573 frames) paired with one teleoperation demonstration. The independently measured preparation stages sum to $a = 2674$ s, dominated by object

segmentation; each pairing then takes $b = 347$ s, dominated by rendering and compositing. With the preparation reused across M demonstrations, the cost per generated demonstration is $a/M + b$, or 882 s for $M = 5$. This time is computation that runs without an operator or a robot. The inputs are what occupy people and hardware: a human episode needs no robot and is shorter to record with fewer people, whereas every teleoperation demonstration occupies an operator and the robot for its whole duration.

Table 8. Wall-clock time per stage, in seconds. Human-episode stages run once per episode; pairing stages run once per (episode, demonstration) pair.

Stage	Per human episode	Per pair
Object masks	1947.3	
Hand trajectory recovery	122.0	
Provisional hand-root path	6.8	
Visible human-hand masks	172.0	
Human removal (inpainting)	425.8	
Temporal alignment (PHTA) and teleoperation preparation		48.9
Hand-root anchoring and arm inverse kinematics		8.7
Output scheduling		0.1
Visibility transfer, rendering, and compositing		289.4
Total	$a = 2673.8$	$b = 347.2$
Per generated demonstration, $M = 5$	$a/M + b = 881.9$	

One sweeping episode (573 frames at 30 Hz) paired with a 706-sample teleoperation demonstration, producing 489 output frames at 15 Hz, on a host with two NVIDIA RTX A5000 GPUs and an Intel Xeon Silver 4309Y CPU. Object segmentation and hand recovery ran on different GPUs at the same time, so the values are full-stage wall-clock times under shared resources rather than dedicated-device throughput. Totals are computed from unrounded stage times.

5. Discussion

5.1. Interpretation

The pipeline assigns complementary roles to the two sources: human video supplies the scene and hand-root trajectory, while teleoperation supplies the finger position-target path. Temporal composition uses hand-state transitions and normalized translation context to establish when to apply that path. Visual composition transfers the front-or-behind relation between hand parts and the object. The behavior cloning results follow this division. The composed data turn five teleoperation demonstrations, which alone do not yield a working policy on two of the four tasks, into teleoperation-level data that match one hundred demonstrations on two tasks. Deriving the finger commands from the human hand instead lowers the success rate on every task, which is the executability gap the design is built around. The gap to the larger teleoperation budget is largest on pouring, the task in which the cup is tilted to pour, on which the temporal alignment is also less accurate than on the apple task (Table 7).

5.2. Limitations

Alignment. On the pouring task the baseline is more accurate at the completion of hand opening, and PHTA still produces isolated errors of several seconds. The complete-target assignment assumes a source counterpart for each selected teleoperation event and does not reject a match solely for a low score. Translation progress becomes less informative when paths differ substantially or reconstruction jitter dominates accumulated motion; the current implementation requires nonzero translation.

Physical execution. The transferred finger targets are combined with a different hand-root trajectory. Contact behavior and task success depend on this combination and the physical scene; the policies were evaluated on four tasks with one robot, and pouring, which involves tilting the cup, retains the largest gap to the larger teleoperation budget.

Evaluation conditions. The objects were placed at nearly the same initial position in every trial, so generalization to other object placements is not evaluated. The robot is rendered from its URDF meshes without adapting its appearance to the real robot.

The camera is stationary. No camera motion is estimated: the hand trajectory, the depth terms, and the compositing are all expressed in one fixed camera frame. Handling a moving camera would require camera tracking, which the pipeline does not include.

The two sources must describe the same task. Composition assumes the human episode and the teleoperation demonstration perform the same manipulation on the same object. Recombining across tasks or across objects is outside the present scope, and the alignment carries no mechanism to detect such a mismatch.

6. Conclusions

We presented a pipeline that composes independently recorded human RGB-D episodes and robot teleoperation logs. Human video supplies the scene and metric hand-root trajectory; teleoperation supplies the robot finger position targets. PHTA associates ordered hand-state transitions using deformation evidence and translation-progress context, then transfers the complete command vector through a shared temporal correspondence. The human hand is replaced by a robot render under an occlusion relation read from the source video, without requiring an object mesh or pose. Compatible source–teleoperation pairings expand the composed dataset while reusing perception results across pairings. The composed data reach teleoperation quality: on four real-robot tasks, five teleoperation demonstrations mixed with twenty human episodes train policies that match one hundred teleoperation demonstrations on two tasks and come within one and four trials of them on the other two.

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Abbreviations

The following abbreviations are used in this manuscript:

BIC	Bayesian information criterion
CAD	Computer-aided design
DTW	Dynamic time warping
MANO	Hand model with articulated and non-rigid deformations
PHTA	Persistent hand-state transition alignment
RGB-D	Color and depth
SLAM	Simultaneous localization and mapping
URDF	Unified Robot Description Format
vMF	von Mises–Fisher

Appendix A. Implementation Details

Appendix A.1. Calibration and Per-Pair Computation

The source and target log-energy mixtures and the direction concentration are loaded as fixed parameters. Event extraction, target-inventory selection, ordered association, and local timing measures are computed for each supplied pair.

Appendix A.2. Branch Direction and Trajectory Support

The branch center is the energy-weighted average of its scale-node centers. If the sum of deformation vectors is numerically zero, the largest-energy node supplies the direction. Progress-support intervals narrower than one canonical cell are expanded to that resolution, and a positive numerical floor is applied before the logarithm. The progress coordinate requires nonzero translation.

Appendix A.3. Local Timing Support

For fewer than three positive energy atoms in a source cell, the implementation retains the atoms directly. A cell with no positive projected increments receives a narrow timing atom at its event center. Insufficient source candidates and numerical failures are reported explicitly.

Appendix A.4. Correspondence Knots and Output

Cell boundaries, cumulative-mass breakpoints, and native target command samples are retained in the dense correspondence. After root fitting and inverse kinematics, the merged source and command supports define the path sampled for fixed-rate output. The continuous interval durations are summed and the complete trajectory is sampled once at the output rate; only the total is rounded, with the sub-frame remainder distributed along the path. Robot poses are interpolated on this continuous time coordinate while the corresponding source RGB index advances monotonically and may repeat.

Appendix A.5. Hand-Root Fit and Output Checks

The global source-root fit uses a fixed regularization weight of 64, applied to translation and to a continuous quaternion representation separately. Here y is the original root trajectory and D^2 is the second-difference operator:

$$\min_x \ \|x - y\|^2 + 64\,\|D^2 x\|^2. \tag{A1}$$

Each output row carries six right-arm joint positions and fifteen right-hand position-target commands in radians, together with the index of the source RGB frame it is drawn from.

Each output is checked for a nondecreasing source index and for commanded joint velocities within the limits of Section 3.5.

Appendix B. Alignment Errors per Boundary

Table A1. Alignment error for each hand boundary, in seconds: median / 90th percentile / maximum.

Task	Boundary	Boundaries	PHTA (ours)	Pairwise DTW
Apple in cup	closing start	147	0.032 / 0.069 / 0.104	1.167 / 2.147 / 5.100
	closing completion	147	0.116 / 0.188 / 0.249	0.967 / 1.967 / 4.967
	opening start	126	0.038 / 0.070 / 0.105	3.050 / 4.050 / 5.233
	opening completion	126	0.141 / 0.296 / 0.517	2.917 / 3.983 / 5.167
Cup pouring	closing start	55	0.488 / 1.857 / 5.844	4.333 / 8.867 / 10.233
	closing completion	125	0.993 / 1.956 / 10.491	2.767 / 8.160 / 9.567
	opening start	125	0.044 / 0.429 / 6.252	0.867 / 1.540 / 2.300
	opening completion	100	0.452 / 1.242 / 6.688	0.200 / 0.443 / 1.300

References

1. Chen, H.; Dong, T.; Wu, T.; Wang, L.; Jangir, Y.; Niu, Y.; Ye, Y.; Bharadhwaj, H.; Erickson, Z.; Ichnowski, J. Dexterous Manipulation Policies from RGB Human Videos via 3D Hand-Object Trajectory Reconstruction. *arXiv* **2026**, *arXiv:2602.09013*.
2. Zhang, G.; Xu, Q.; Zhang, H.; Ma, J.; He, L.; Bao, Y.; Ping, Z.; Yuan, Z.; Lu, C.; Yuan, C.; et al. UniDex: A Robot Foundation Suite for Universal Dexterous Hand Control from Egocentric Human Videos. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026.
3. Xu, M.; Zhang, H.; Hou, Y.; Xu, Z.; Fan, L.; Veloso, M.; Song, S. DexUMI: Using Human Hand as the Universal Manipulation Interface for Dexterous Manipulation. In Proceedings of the Conference on Robot Learning (CoRL), 2025.
4. Romero, J.; Tzionas, D.; Black, M.J. Embodied Hands: Modeling and Capturing Hands and Bodies Together. *ACM Trans. Graph.* **2017**, *36*, 245.
5. Qin, Y.; Wu, Y.H.; Liu, S.; Jiang, H.; Yang, R.; Fu, Y.; Wang, X. DexMV: Imitation Learning for Dexterous Manipulation from Human Videos. In Proceedings of the European Conference on Computer Vision (ECCV), 2022.
6. Shaw, K.; Bahl, S.; Pathak, D. VideoDex: Learning Dexterity from Internet Videos. In Proceedings of the Conference on Robot Learning (CoRL), 2022.
7. Wang, C.; Shi, H.; Wang, W.; Zhang, R.; Fei-Fei, L.; Liu, C.K. DexCap: Scalable and Portable Mocap Data Collection System for Dexterous Manipulation. In Proceedings of the Robotics: Science and Systems (RSS), 2024.
8. Liu, Y.; Shin, W.C.; Han, Y.; Chen, Z.; Ravichandar, H.; Xu, D. ImMimic: Cross-Domain Imitation from Human Videos via Mapping and Interpolation. In Proceedings of the Conference on Robot Learning (CoRL), 2025.
9. Kareer, S.; Patel, D.; Punamiya, R.; Mathur, P.; Cheng, S.; Wang, C.; Hoffman, J.; Xu, D. EgoMimic: Scaling Imitation Learning via Egocentric Video. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2025, pp. 13226–13233.
10. Lepert, M.; Fang, J.; Bohg, J. Phantom: Training Robots Without Robots Using Only Human Videos. In Proceedings of the Conference on Robot Learning (CoRL), 2025.
11. Lepert, M.; Fang, J.; Bohg, J. Masquerade: Learning from In-the-wild Human Videos using Data-Editing. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2026.
12. Dessalene, E.; Mantripragada, P.; Maynord, M.; Aloimonos, Y. EmbodiSwap for Zero-Shot Robot Imitation Learning. *arXiv* **2025**, *arXiv:2510.03706*.
13. Ci, H.; Liu, X.; Yang, P.; Song, Y.; Shou, M.Z. H2R-Grounder: A Paired-Data-Free Paradigm for Translating Human Interaction Videos into Physically Grounded Robot Videos. *arXiv* **2025**, *arXiv:2512.09406*.
14. Zhang, T.; Xia, S.; Wang, Y.; Jin, Q. EasyMimic: A Low-Cost Framework for Robot Imitation Learning from Human Videos. In Proceedings of the IEEE International Conference on Robotics and Automation (ICRA), 2026.
15. Chen, L.Y.; Xu, C.; Dharmarajan, K.; Irshad, M.Z.; Cheng, R.; Keutzer, K.; Tomizuka, M.; Vuong, Q.; Goldberg, K. RoVi-Aug: Robot and Viewpoint Augmentation for Cross-Embodiment Robot Learning. In Proceedings of the Conference on Robot Learning (CoRL), 2024.
16. Qin, Y.; Yang, W.; Huang, B.; Van Wyk, K.; Su, H.; Wang, X.; Chao, Y.W.; Fox, D. AnyTeleop: A General Vision-Based Dexterous Robot Arm-Hand Teleoperation System. In Proceedings of the Robotics: Science and Systems (RSS), 2023.

17. Yu, Z.; Zafeiriou, S.; Birdal, T. Dyn-HaMR: Recovering 4D Interacting Hand Motion from a Dynamic Camera. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
18. Pavlakos, G.; Shan, D.; Radosavovic, I.; Kanazawa, A.; Fouhey, D.; Malik, J. Reconstructing Hands in 3D with Transformers. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
19. Teed, Z.; Deng, J. DROID-SLAM: Deep Visual SLAM for Monocular, Stereo, and RGB-D Cameras. In Proceedings of the 35th Conference on Neural Information Processing Systems (NeurIPS), 2021.
20. Duran, E.; Kocabas, M.; Choutas, V.; Fan, Z.; Black, M.J. HMP: Hand Motion Priors for Pose and Shape Estimation from Video. In Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV), 2024, pp. 6353–6363.
21. Kirillov, A.; Mintun, E.; Ravi, N.; Mao, H.; Rolland, C.; Gustafson, L.; Xiao, T.; Whitehead, S.; Berg, A.C.; Lo, W.Y.; et al. Segment Anything. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023, pp. 4015–4026.
22. Carion, N.; Gustafson, L.; Hu, Y.T.; Debnath, S.; Hu, R.; Suris, D.; Ryali, C.; Alwala, K.V.; Khedr, H.; Huang, A.; et al. SAM 3: Segment Anything with Concepts. In Proceedings of the International Conference on Learning Representations (ICLR), 2026.
23. Chen, K.; Ramanan, D.; Khurana, T. Using Diffusion Priors for Video Amodal Segmentation. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
24. Zhou, S.; Li, C.; Chan, K.C.K.; Loy, C.C. ProPainter: Improving Propagation and Transformer for Video Inpainting. In Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV), 2023.
25. Wen, B.; Yang, W.; Kautz, J.; Birchfield, S. FoundationPose: Unified 6D Pose Estimation and Tracking of Novel Objects. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
26. Lindeberg, T. Feature Detection with Automatic Scale Selection. *Int. J. Comput. Vis.* **1998**, *30*, 79–116.
27. Edelsbrunner, H.; Letscher, D.; Zomorodian, A. Topological Persistence and Simplification. *Discrete Comput. Geom.* **2002**, *28*, 511–533.
28. Schwarz, G. Estimating the Dimension of a Model. *Ann. Stat.* **1978**, *6*, 461–464.
29. Mardia, K.V.; Jupp, P.E. *Directional Statistics*; Wiley: Chichester, UK, 2000.
30. Barlow, R.E.; Bartholomew, D.J.; Bremner, J.M.; Brunk, H.D. *Statistical Inference Under Order Restrictions*; Wiley: London, UK, 1972.
31. Sakoe, H.; Chiba, S. Dynamic Programming Algorithm Optimization for Spoken Word Recognition. *IEEE Trans. Acoust. Speech Signal Process.* **1978**, *26*, 43–49.
32. Cuturi, M.; Blondel, M. Soft-DTW: a Differentiable Loss Function for Time-Series. In Proceedings of the International Conference on Machine Learning (ICML), 2017.
33. Dwibedi, D.; Aytar, Y.; Tompson, J.; Sermanet, P.; Zisserman, A. Temporal Cycle-Consistency Learning. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2019.
34. Mandlekar, A.; Nasiriany, S.; Wen, B.; Akinola, I.; Narang, Y.; Fan, L.; Zhu, Y.; Fox, D. MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations. In Proceedings of the Conference on Robot Learning (CoRL), 2023.
35. Casiez, G.; Roussel, N.; Vogel, D. 1€ Filter: A Simple Speed-based Low-pass Filter for Noisy Input in Interactive Systems. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI), 2012, pp. 2527–2530.
36. Kabsch, W. A solution for the best rotation to relate two sets of vectors. *Acta Crystallogr. Sect. A* **1976**, *32*, 922–923.
37. Chi, C.; Xu, Z.; Feng, S.; Cousineau, E.; Du, Y.; Burchfiel, B.; Tedrake, R.; Song, S. Diffusion Policy: Visuomotor Policy Learning via Action Diffusion. In Proceedings of the Robotics: Science and Systems (RSS), 2023.
38. Lipman, Y.; Chen, R.T.Q.; Ben-Hamu, H.; Nickel, M.; Le, M. Flow Matching for Generative Modeling. In Proceedings of the International Conference on Learning Representations (ICLR), 2023.

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